

Yeast Miniprep: Zymolyase Digestion

Model organism *Saccharomyces cerevisiae* has a thick **chitin cell wall**, preventing simple extraction of the plasmid. This miniprep breaks down the Chitin walls, and extracts **2 μ -based plasmids** (type of plasmid high in copy numbers), yielding approximately **0.01-0.3ng** of plasmid per **1.5mL** of overnight culture. The plasmid DNA is recovered in a TE buffer, and is able to be used in *E.coli* transformations, western blotting, PCR, etc.

Reagents

Notes:

- This is the amount of reagent used per 1.5mL of yeast culture. The kit is able to complete approximately 100x yeast minipreps.

✓	Name	Quantity	Location/Notes
☐	Yeast overnight culture	1.5mL/0.01-0.3ng	Starting material
☐	Digestion buffer	150 μ L	From kit
☐	Lysis buffer	150 μ L	From kit
☐	Neutralizing buffer	150 μ L	From kit
☐	Zymolyase and Storage Buffer	2 μ L	From kit Store at -20°C
☐	Isopropanol (2-Propanol)	400 μ L	
☐	TE buffer	35 μ L	

Equipment

✓	Name	Quantity	Location/Notes
☐	Microcentrifuge		
☐	Incubator Shaker		
☐	Micropipette		
☐	Vortex Mixer (Optional)		
☐	Micrientrifuge tubes (1.5mL)		

Instructions

Notes:

- These instructions are allocated for only *one* sample of **1.5mL**.
- If you wish to process multiple samples, **DO NOT multiply** the reagent and culture volume. It will be hard to centrifuge volumes larger than 1.5mL, and resuspension of the pellet will be difficult.
- Separate into ***different test tubes*** when processing more than 1.5mL of culture, and complete the steps as stated.

Preparing the Reagents

1. Add 200 μ L of the supplied **Storage buffer** to lyophilized (freeze dried) **Zymolase** enzyme.
 - a. The freeze dried Zymolase enzyme should be a *powder*
2. Microcentrifuge to mix and dissolve the enzyme completely into the liquid buffer
3. Store the newly dissolved Zymolase at -20°C.

Growing the Yeast

1. Grow yeast at 30°C in selected medium

Protocol

1. Micropipette 0.5-1.0mL of full grown yeast cells into 1.5mL microcentrifuge tubes.
2. Centrifuge at 600 x g for 2 minutes

3. Discard supernatant (top layer of liquid)
 - a. Note: Using a micropipette, *carefully* extract the layer of liquid and discard it. **DO NOT** micropipette and discard the pellet at the bottom of the microcentrifuge tube.
4. Micropipette **150µl of digestion buffer**, and add it to the pellet.
5. Micropipette **2µl of Zymolase** enzyme reagent (previously prepared), add it to the pellet and digestion buffer solution.
6. Close the lid of the tube, and vortex or flick the solution to **resuspend the pellet** (mixed back into the liquid)
7. Incubate at **37°C for 15-60 minutes**
 - a. minimum incubation time of 15 minutes
 - b. Longer incubation times are optional
8. Micropipette **150µl of lysis buffer**, and add it to the previously incubated tube. Vortex to mix well.
9. Micropipette **150µl of neutralizing buffer**, and add it to the tube. Vortex to mix well.
10. Centrifuge at MAXIMUM speed for **2 minutes**.
11. **Transfer supernatant** to new tubes.
 - a. DO NOT throw out the liquid like in the previous step.
12. Add **400µl isopropanol** (2-propanol) to the tube. Vortex to mix well.
13. Centrifuge at MAXIMUM speed for **8 minutes**.
14. Aspirate the supernatant (micropipette to remove the liquid again), and discard it.
 - a. Try your best to not touch the pellet, while removing as much liquid as possible. The pellet is the DNA.
15. Spin the tube in the centrifuge briefly, and aspirate again (removes any leftover supernatant).
16. Add **35 µl of TE buffer** to the plasmid pellet to resuspend it. You do not need to remove/transfer the pellet before adding the TE buffer.

Reagent Tracker (use this table as needed)	Quantity
☐ Yeast overnight culture	0.5-1.0mL
☐ Digestion buffer	150µL
☐ Lysis buffer	150µL
☐ Neutralizing buffer	150µL
☐ Zymolyase and Storage Buffer	2µL

- Kit works up to 23kb

References

https://files.zymoresearch.com/protocols/_d2001_zymoprep_yeast_plasmid_miniprep_i.pdf (From the kit)

<https://www.zymoresearch.com/collections/zymoprep-yeast-plasmid-miniprep/products/zymoprep-yeast-plasmid-miniprep-i> (Zymoprep kit)

https://www.neb.com/en-ca/protocols/high-molecular-weight-dna-hmw-dna-extraction-from-yeast?utm_source=chatgpt.com (From high molecular weight DNA)

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3586270/> (Why we used 2 micron base plasmids)